HYKE Weekly seminar

October 9th

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Today : Finish §2.3

Def
- The uncertainty of an observable A in the state $|\psi\rangle$ is defined as

$$\Delta_\psi(A) = \sqrt{\langle\psi|(A-\langle A\rangle_\psi \mathbb{1})^2\psi\rangle}$$
$$= \sqrt{\langle (A-\langle A\rangle_\psi \mathbb{1})^2\rangle_\psi}$$
$$\overset{\vee}{=} \sqrt{\langle A^2\rangle_\psi - \langle A\rangle_\psi^2}$$

→ Standard deviation of A.

- $\Delta_\psi(A) = 0 \Longrightarrow$ The value of the observable A in the state $|\psi\rangle$ is sharp.

Since $(\Delta_\psi(A))^2 = \langle (A - \langle A\rangle_\psi \mathbb{1})^2\rangle_\psi$ → Scalar valued

$$= \langle A^2 - 2\langle A\rangle_\psi A + \langle A\rangle_\psi^2\rangle_\psi$$
$$= \langle A^2\rangle_\psi - \langle A\rangle_\psi^2$$

$$\boxed{\Delta_\psi(A) = 0.} \iff \langle A^2\rangle_\psi = \langle A\rangle_\psi^2 .$$

Let $|\lambda\rangle$ be an eigen-state with

$$\underline{A|\lambda\rangle = \lambda|\lambda\rangle} . \quad \Lambda : \text{set of eigen values.}$$

$$\Rightarrow \langle A^2\rangle_\psi = \langle\psi|A^2|\psi\rangle \qquad \boxed{\mathbb{1} = \sum_{\lambda\in\Lambda} |\lambda\rangle\langle\lambda|}$$

$$= \sum_{\lambda_1,\lambda_2\in\Lambda} \langle\psi|\lambda_1\rangle \underbrace{\langle\lambda_1|A^2|\lambda_2\rangle}_{\lambda_2^2\langle\lambda_1|\lambda_2\rangle} \langle\lambda_2|\psi\rangle$$

$$= \sum_{\lambda_1,\lambda_2\in\Lambda} \langle\psi|\lambda_1\rangle\, \lambda_2^2 \underbrace{\langle\lambda_1|\lambda_2\rangle}_{\delta_{\lambda_1\lambda_2}} \langle\lambda_2|\psi\rangle . \qquad \boxed{\{|\lambda\rangle\} : \text{ONB}}$$

$$= \sum_{\lambda\in\Lambda} |\langle\psi|\lambda\rangle|^2 \lambda^2 .$$

$$\langle A\rangle_\psi = \langle\psi|A|\psi\rangle = \sum_{\lambda\in\Lambda} |\langle\psi|\lambda\rangle|^2 \lambda .$$

$$\boxed{\langle A^2\rangle_\psi = \langle A\rangle_\psi^2} \Rightarrow \sum_{\lambda\in\Lambda} |\langle\psi|\lambda\rangle|^2\lambda^2 = \left(\sum_{\lambda\in\Lambda} |\langle\psi|\lambda\rangle|^2\lambda\right)^2$$

$$P_\psi(\lambda) = |\langle\psi|\lambda\rangle|^2 \quad (\text{Probability}).$$

Hölder Ineq.

$$\Big(\underbrace{\sum_\lambda P_\psi(\lambda)\lambda^2}_{\langle A^2\rangle_\psi}\Big)\Big(\underbrace{\sum_\lambda P_\psi(\lambda)}_{1}\Big) \geq \left(\sum_\lambda P_\psi(\lambda)|\lambda|\right)^2$$

$$\geq \left(\sum_\lambda P_\psi(\lambda)\lambda\right)^2 = \underline{\langle A\rangle_\psi^2} .$$

Triangle Ineq.

$\Rightarrow$ Now, we want to find condition of equality holds.

Hölder-Inequality

$$\frac{P_\psi(\lambda)\lambda^2}{P_\psi(\lambda)} = \text{const.} \Rightarrow \lambda^2 \text{ should be constant.}$$

Triangle Inequality,

Signs of $\lambda \in \Lambda$ should be same.

$\Rightarrow \exists! \lambda \in \Lambda$ s.t. $P_\psi(\lambda) \neq 0$.

$\Rightarrow$ $\boxed{\psi \text{ is an eigen state of } A.}$

$\rightarrow$ Result of $\boxed{\text{Proposition 2.17.}}$

Def Two observables A, B are called

$$\begin{cases} \text{compatible if } [A,B]=0. \\ \text{incompatible if } [A,B]\neq 0. \end{cases}$$

Prop. (Uncertainty Principle)

self-adjoint operators

For any observables $A, B \in B_{s\text{-}a}(H)$ and state $|\psi\rangle \in H$ the following uncertainty relation holds.

$$\Delta_\psi(A)\cdot\Delta_\psi(B) \geq \left|\left\langle \frac{1}{2i}[A,B]\right\rangle_\psi\right|.$$

$\boxed{\text{Idea of proof} \to \text{Just Hölder inequality.}}$

E.g. $A = \hat{x}$, $B = \hat{p}$.

Since $[\hat{x}, \hat{p}] = i\hbar$, we have

$$\boxed{\Delta_\psi(\hat{x}) \cdot \Delta_\psi(\hat{p}) \geq \left| \left\langle \frac{1}{2i} \cdot i\hbar \right\rangle_\psi \right| = \frac{\hbar}{2}}$$

$\frac{\hbar}{2}$

position momentum,

$\Rightarrow$ We cannot measure position and momentum simultaneously.

$\boxed{\text{Postulate 3.}}$ (Projection Postulate)

If a measurement of the observable A on a quantum mechanical system in the pure state $|\Psi\rangle \in \mathcal{H}$ yields the eigenvalue λ, then the measurement has affected the following state projection.

collapse of the wave function.

projection

$$|\Psi\rangle \xrightarrow[\boxed{\hat{A} \to \lambda}]{\text{measurement}} \frac{P_\lambda |\psi\rangle}{\| P_\lambda |\psi\rangle \|}$$

normalization constant.

(Every quantum state satisfies $\|\psi\|_2 = 1$)

Let $\underbrace{\{\lambda \in \Lambda\}}_{\boxed{\text{ONB}}}$ s.t $\underline{\hat{A}|\lambda\rangle = \lambda|\lambda\rangle}$

If we set

$$|\Psi\rangle = \sum_{\lambda\in\Lambda} C_\lambda |\lambda\rangle \xrightarrow[\hat{A}\to\lambda_0]{\text{measurement}} C|\lambda_0\rangle.$$

$|C| = 1$ complex number $\boxed{C = e^{i\phi}}$

$\boxed{(C_{\lambda_0} \neq 0)}$

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* Time evolution.

$\mathcal{U}(t_2,t_1)$ is a time-evolution operator:

$$|\psi(t_1)\rangle \xrightarrow{\text{time-evolution.}} |\psi(t_2)\rangle = \mathcal{U}(t_2,t_1)|\psi(t_1)\rangle$$

(No measurement)

Prop. ① $\mathcal{U}(t_2,t_1)$ should be unitary. $\forall\ t_1 < t_2$.

② $\begin{cases} \boxed{i\dfrac{d}{dt}\mathcal{U}(t,t_0) = H(t)\,\mathcal{U}(t,t_0)} \\ \boxed{\mathcal{U}(t_0,t_0) = \mathbb{1}} \end{cases}$ $\quad \underline{(H(t) \in B_{s.a}(\mathcal{H}))}$.

$\dot{\mathcal{U}}(t,t_0) = -i\,H(t)\,\mathcal{U}(t,t_0)$

$H(t) = H^*(t)$

(pf). ① Since $\||\psi(t_1)\rangle\| = \||\psi(t_2)\rangle\| = 1$,

$\mathcal{U}(t_2,t_1)$ should be unitary.

② $\dfrac{d}{dt}(\mathcal{U}^*(t,t_0)\,\mathcal{U}(t,t_0)) = \boxed{\dot{\mathcal{U}}^*(t,t_0)}\,\mathcal{U}(t,t_0) + \mathcal{U}^*(t,t_0)\,\dot{\mathcal{U}}(t,t_0)$

$= (i\,\mathcal{U}^*(t,t_0)\,H^*(t))\,\mathcal{U}(t,t_0) - \mathcal{U}^*(t,t_0)\,(i\,H(t)\,\mathcal{U}(t,t_0))$

$= 0.$

$$\underline{\mathcal{U}^{\dagger}(t,t_0)\,\mathcal{U}(t,t_0)} = \mathcal{U}^{\dagger}(t_0,t_0)\,\mathcal{U}(t_0,t_0)$$
$$= \mathbb{1}\cdot\mathbb{1} = \mathbb{1}.$$

$\Longrightarrow \mathcal{U}(t,t_0)$ is an unitary operator. $\square$

Postulate 4 (Time-evolution)

$|\psi(t_0)\rangle$: State at time t_0

$\xrightarrow[\text{No measurement}]{}$ $|\psi(t)\rangle = \mathcal{U}(t,t_0)\,|\psi(t_0)\rangle$
: State at time t.

Recall that Schrödinger Eq

$$i\,\partial_t|\psi(t)\rangle = H(t)\,|\psi(t)\rangle$$

$$\Rightarrow i\,\underline{\partial_t}\big(\mathcal{U}(t,t_0)\,|\psi(t_0)\rangle\big) = H(t)\,\mathcal{U}(t,t_0)\,|\psi(t_0)\rangle$$

$\hookrightarrow$ does not depend on "t"

$$\Rightarrow \underline{\big(i\,\partial_t\,\mathcal{U}(t,t_0)\big)}\,\boxed{|\psi(t_0)\rangle} = \underline{\big(H(t)\,\mathcal{U}(t,t_0)\big)}\,\boxed{|\psi(t_0)\rangle}.$$

$$\Rightarrow \boxed{i\,\partial_t\,\mathcal{U}(t,t_0) = H(t)\,\mathcal{U}(t,t_0).}$$

$\boxed{\text{Spin}}$

- Important concept in Q.M.
- Each particle (electron, proton, photon, neutron, ...) has a different spin structure.
- In Quantum Computation, we use "electron"

So in this book, we'll study about spin of electron.

In classical computer ⇒ $\boxed{\text{0 or 1. } \quad \boxed{\begin{matrix}N\\S\end{matrix}} \text{ or } \boxed{\begin{matrix}S\\N\end{matrix}}}$ → $\boxed{\text{bit}}$

In Quantum computer ⇒ $\boxed{\text{Qubit}}$ → $\boxed{\text{Express a state from electron.}}$

Spin is abstract concept, but physical meaning of spin is similar to angular momentum of quantum particles.

We'll study angular momentum in Q.M.

In classical mechanics,

$$\boxed{\vec{L} = \vec{x} \times \vec{p}}$$ Cross product.

even perm of (1,2,3)

Levi-Civita Symbol.

$$L_i = (\vec{x}\times\vec{p})_i = \varepsilon_{ijk}\, x_j\, p_k$$

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \\ -1 & \text{if } (i,j,k) \\ 0 & \text{if } \exists \text{ repeated index} \end{cases}$$

odd perm.

$\varepsilon_{113} = 0$

($1=x$, $2=y$, $3=z$

L_z : z-component of angular momentum vector $\vec{L}$)

$$\begin{cases} L_x = y p_z - z p_y \\ L_y = z p_x - x p_z \\ L_z = x p_y - y p_x. \end{cases} \Rightarrow \begin{pmatrix} \hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \\ \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \\ \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \end{pmatrix}$$

$$\boxed{\hat{L}_i = \varepsilon_{ijk}\, \hat{x}_j\, \hat{p}_k}$$

($L_1 = L_x$, $L_2 = L_y$, $L_3 = L_z$

$x_1 = x$, $x_2 = y$, $x_3 = z$.

$p_1 = p_x$, $p_2 = p_y$, $p_3 = p_z$.)

$[x_i, x_j] = 0$

$[p_i, p_j] = 0$

$[x_i, p_j] = i\hbar\,\delta_{ij}$

if $i=j \Rightarrow \underbrace{[x, p_x]}_{= i\hbar} = [y, p_y] = [z, p_z]$ $= i\hbar$

If $i \neq j \Rightarrow [x, p_y] = 0$.

Einstein Notation.

tensor contraction

$$\varepsilon_{ijk}\,\varepsilon_{ilm} = \underbrace{\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}}_{\text{rank-4 tensor}}$$: tensor - Identity.

ε_{ijk} : rank-3 tensor.

$3+3-2=4$

$j, l, k, m \in \{1,2,3\}$

$\Rightarrow \boxed{3^4 = 81}$

$$[\hat{L}_i, \hat{L}_j] = [\varepsilon_{ilm}\hat{x}_l\hat{p}_m, \varepsilon_{jkn}\hat{x}_k\hat{p}_n]$$

$$= \varepsilon_{ilm}\varepsilon_{jk\cancel{m}n}[\hat{x}_l\hat{p}_m, \hat{x}_k\hat{p}_n]$$

$$= (\quad " \quad)(\hat{x}_l\hat{p}_m\hat{x}_k\hat{p}_n - \hat{x}_k\hat{p}_n\hat{x}_l\hat{p}_m)$$

$$\hat{x}_l\underbrace{[\hat{p}_m, \hat{x}_k]}_{-i\hbar\delta_{mk}}\hat{p}_n + \hat{x}_l\hat{x}_k\hat{p}_m\hat{p}_n$$

$$-\hat{x}_k\underbrace{[\hat{p}_n, \hat{x}_l]}_{-i\hbar\delta_{nl}}\hat{p}_m - \hat{x}_k\hat{x}_l\hat{p}_n\hat{p}_m$$

$$= 0 \left(\because [\hat{x}_l, \hat{x}_k] = 0,\ [\hat{p}_m, \hat{p}_n] = 0\right)$$

$$= \boxed{\varepsilon_{ilm}\varepsilon_{jk\cancel{m}n}}\left(i\hbar\delta_{nl}\hat{x}_k\hat{p}_m - i\hbar\delta_{mk}\hat{x}_l\hat{p}_n\right)$$

$$= i\hbar\left(\varepsilon_{ilm}\varepsilon_{jkl}\hat{x}_k\hat{p}_m - \varepsilon_{ilk}\varepsilon_{jkn}\hat{x}_l\hat{p}_n\right)$$

$$= i\hbar\left(\boxed{\varepsilon_{lmi}\varepsilon_{ljk}}\hat{x}_k\hat{p}_m + \boxed{\varepsilon_{kil}\varepsilon_{kjn}}\hat{x}_l\hat{p}_n\right)$$

$$\delta_{mj}\delta_{ik} - \delta_{mk}\delta_{ij} \qquad \delta_{ij}\delta_{ln} - \delta_{in}\delta_{jl}$$

$$= \cdots$$

$$= i\hbar(\hat{x}_i\hat{p}_j - \hat{x}_j\hat{p}_i) = i\hbar\varepsilon_{ijk}\hat{L}_k.$$

$$\therefore [\hat{L}_i, \hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k.$$

Relation between Angular momentum operator.

Now, come back to "Spin".

$\boxed{\text{Def}}$. The matrices $\sigma_j \in Mat_{2\times2}(\mathbb{C})$ indexed by either $\boxed{\hat{j}} \in \{1,2,3\}$ or $\boxed{\hat{j}} \in \{x,y,z\}$ and defined as

$$\sigma_x = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\Rightarrow$ Pauli-matrices. $\left(\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$.

$\{\sigma_i\} = \{\sigma_1, \sigma_2, \sigma_3\}$.

$\{\sigma_\alpha\} = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$

- $\hat{S}_j = \frac{1}{2}\sigma_j$: spin operator for electron.

From the simple calculation, we have

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k.$$

$$\text{S}|$$

$$[\hat{L}_i, \hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k$$

$$L_i = \frac{2}{\hbar}\sigma_i$$

We use Pauli-Matrices to express spin operator.

$$|\uparrow_{\hat z}\rangle = |0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$$

$$|\downarrow_{\hat z}\rangle = |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

$$\Longrightarrow \left\{ \begin{aligned} \hat{S}_z|\uparrow_{\hat z}\rangle &= \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{2}|\uparrow_{\hat z}\rangle \\ \hat{S}_z|\downarrow_{\hat z}\rangle &= -\frac{1}{2}|\downarrow_{\hat z}\rangle. \end{aligned} \right.$$

$|\uparrow_{\hat z}\rangle, |\downarrow_{\hat z}\rangle \Rightarrow$ <u>eigen state of $\hat{S}_z$.</u> $\frac{1}{2}, -\frac{1}{2}$, resp.

Remark.

Since $[\sigma_i, \sigma_j] = i\varepsilon_{ijk}\sigma_k \neq 0$ for $i \neq j$,

σ_i, σ_j are Incompatible.

$\Longleftrightarrow \hat{S}_i, \hat{S}_j$ are Incompatible $\forall i \neq j$.

2.3.2 Mixed State.

<u>Overview</u>

- pure state : $|\psi\rangle \in \mathbb{H}$ with $\|\psi\| = 1$.
- mixed state : $\rho \in B_{s.a}(\mathbb{H})$ with $\begin{pmatrix} tr(\rho) = 1. \\ \rho \geq 0. \\ \vdots \end{pmatrix}$

density operator.

$$\rho = \sum_{\alpha \in \Lambda} p_\alpha |\psi_\alpha\rangle\langle\psi_\alpha|$$

with $\sum_\alpha p_\alpha = 1$, $\{|\psi_\alpha\rangle\}$: ONB.

Remark. Density operator for pure state $|\psi\rangle$

$$\Longrightarrow \boxed{\rho = |\psi\rangle\langle\psi|.}$$

Postulate 5 (Mixed State)

ρ : density operator satisfies following properties.

① ρ : self-adjoint $\rho^* = \rho$.

② positive $\boxed{\rho \geq 0.}$ $(\langle \psi | \rho | \psi \rangle \geq 0.\ \forall |\psi\rangle \in \mathbb{H})$

③ ρ has trace 1. $\mathrm{tr}(\rho) = 1$.

$\Rightarrow D(\mathbb{H})$: set of density operators defined on $\mathbb{H}$.

Remark.

convex combination.

① $\rho_1, \rho_2 \in D(\mathbb{H}) \Rightarrow \boxed{\lambda\rho_1 + (1-\lambda)\rho_2} \in D(\mathbb{H})\ (0 \leq \lambda \leq 1)$

② $\rho \in D(\mathbb{H}) \Rightarrow U\rho U^* \in D(\mathbb{H})$. $\forall$ unitary U.

Postulate 6.

(pure state)

(just generalization of $\boxed{\text{Postulate } 1-4}$)

$\boxed{\text{Thm 2.24}}$ $\rho \in D(\mathbb{H})$

① $\exists\, p_j$, ONB $\{|\psi_j\rangle\}$ s.t

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|.$$

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② $0 \leq \rho^2 \leq \rho$.

③ $\|\rho\|_{op} \leq 1$.

(pf) ① omit. (Infinite dim — linear algebra).

② $\boxed{\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|}$

$$0 \leq \rho^2 = \sum_{j,k} \left(p_j |\psi_j\rangle\langle\psi_j|\right)\left(p_k |\psi_k\rangle\langle\psi_k|\right)$$

δ_{jk}

$$= \sum_{j,k} p_j p_k |\psi_j\rangle \langle\psi_j|\psi_k\rangle \langle\psi_k| .$$

$$= \boxed{\sum_j p_j^2 |\psi_j\rangle\langle\psi_j| .}$$

$$\leq \sum_j p_j |\psi_j\rangle\langle\psi_j| = \rho .$$

③ $\|\rho\|_{op}^2 = \sup_{\||\psi\rangle\|=1} \left| \rho|\psi\rangle \right|$

$$\left|\sum_j p_j |\psi_j\rangle\langle\psi_j|\psi\rangle\right|^2 \leq \sum p_j$$

Mistake

Trivial.

Prop. 2.25 / 2.26.

$\rho \in D(\mathbb{H})$ Is a pure state $\Longleftrightarrow$ $\rho = \rho^2$.

($\Rightarrow$) Easy. $\||\psi\rangle\|=1$ for pure state

Let $\rho = |\psi\rangle\langle\psi| \Rightarrow \rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi|$

$$= |\psi\rangle\langle\psi| = \rho .$$

$(\Leftarrow)$ $\rho^2 = \sum_j p_j^2 |\psi_j\rangle\langle\psi_j|$

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$$

If $\rho^2 = \rho \Rightarrow$ Matrix representation of ρ^2, ρ should be same.

$\boxed{p_j = 0 \text{ or } 1}$

$\Rightarrow p_j^2 = p_j$ / $\underline{\sum_j p_j = 1}$

$\Rightarrow$ only one $\tau \in \Lambda$ s.t. $p_\tau = 1$.

$\forall j \neq \tau$ s.t. $p_j = 0$.

$\Rightarrow \rho = p_\tau |\psi_\tau\rangle\langle\psi_\tau| = |\psi_\tau\rangle\langle\psi_\tau|$. $\Rightarrow$ $\boxed{\text{pure state}}$

Prop 2.27. Let $\mathbb{H}$ be a finite-dimensional Hilbert space and $\rho \in D(\mathbb{H})$ be a density operator with diagonal form

$\rho = \sum_{j=1}^{n} p_j |\psi_j\rangle\langle\psi_j|$ where $p_j \in (0, 1]$ $\forall j = \{1, \cdots, n\}$

with $n \leq \dim(\mathbb{H})$ are the nonzero eigenvalues of ρ.

$\{|\psi_j\rangle\}$: ONB of its eigenvectors.

Moreover, $\forall \tau \in \{1 \cdots, m\}$ with $m \leq \dim(\mathbb{H})$, $q_i \in (0, 1]$

$\sum_{\tau=1}^{m} q_\tau = 1$, $|\varphi_\tau\rangle \in \mathbb{H}$, $\|\varphi_\tau\| = 1$. Then

$$\boxed{\rho = \sum_{\tau=1}^{m} q_\tau |\varphi_\tau\rangle\langle\varphi_\tau|} \iff \left(\begin{array}{l} m \geq n, \ \exists\, \mathcal{U} \in \mathcal{U}(m) \\ \sqrt{q_\tau} |\varphi_\tau\rangle = \sum_{j=1}^{n} \mathcal{U}_{\tau j} \sqrt{p_j} |\psi_j\rangle \\ \forall \tau = 1 \sim m \end{array} \right)$$

(pf) Omit. (Linear Algebra)